Reference-Conditioned Structural Assessment of Low-Resource Chinese Calligraphy with Overlapping Stroke Parsing and Human-Validated Spatial Scoring

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Abstract

Purpose Calligraphy practice systems must expose local stroke structure, suppress harmless placement variation, and avoid presenting an opaque score as expert aesthetic judgment. We study this problem in a low-resource direct-digital-writing setting.

Methods We represent each character by five overlapping direction masks and one keypoint mask. U-Net, DeepLabV3+, and SegFormer-B2 are evaluated with three seeds on quality-controlled standard and character-disjoint splits. A constrained similarity registration is assessed on 200 real references by controlled perturbations and paired ablations. We introduce aligned spatial-distribution similarity (ASDS), which compares polar occupancy, coarse spatial occupancy, and projection profiles after registration. Score validity is studied with three blinded raters on 150 natural same-character pairs. ASDS is developed post hoc and reserved for a disjoint confirmatory study.

Results DeepLabV3+ obtains $\mathbf{0.9630 \pm 0.0006}$ standard-split Macro Dice, whereas SegFormer-B2 leads character-disjoint evaluation at $\mathbf{0.8866 \pm 0.0008}$. Constrained registration reduces nuisance penalties by 30.118 points (95% CI 29.506–30.727) relative to no alignment while preserving an additional 11.189 points of structural penalty (10.094–12.322). The frozen production and coverage-aware scores correlate with blinded ratings at $\boldsymbol{\rho} = \mathbf{0.297}$ and $\mathbf{0.429}$, respectively. In development-only analysis, ASDS reaches $\boldsymbol{\rho} = \mathbf{0.556}$, with character-grouped out-of-fold $\boldsymbol{\rho} = \mathbf{0.540}$.

Conclusion Overlapping parsing and constrained registration provide auditable structural evidence, while spatial-distribution comparison improves development-stage human association. Independent confirmation is required before ASDS can replace the deployed score, and none of the reported scores should be interpreted as universal calligraphic beauty.

Keywords: Chinese calligraphy, document analysis, multi-label segmentation, reference-based assessment, human validation, explainable feedback

1 Introduction

Digital tools for Chinese calligraphy usually solve one of three tasks: recognizing the written character, identifying a calligraphic style, or generating a visually plausible glyph. A practice system has a different requirement. Given a learner's writing and a designated reference, it should show which structural regions agree, which regions are missing or excessive, and how the global organization differs. The output must be local enough to support revision, yet conservative enough not to masquerade as an expert judgment of brushwork, rhythm, historical authenticity, or beauty.

This requirement creates a document-analysis problem at the interface of segmentation, registration, shape comparison, and human-centred evaluation. First, strokes in a finished raster character intersect and adhere. A pixel at an intersection can legitimately belong to more than one directional structure, making mutually exclusive semantic segmentation inappropriate. Second, direct overlap confounds structural disagreement with global translation, scale, and small rotation. Conversely, an unconstrained warp may erase the very local discrepancy that a teaching system should report. Third, a mathematically convenient overlap score need not follow human perception. The score therefore requires explicit semantic auditing and human validation, not only a high self-consistency result.

The low-resource regime makes the problem harder. Our recovered corpus contains complete six-channel ground truth for only 40 character identities. Writer identity and style provenance cannot be recovered reliably. In this setting, a strong standard-split result can overstate generalization because the same character identities appear during training and testing. We therefore separate ordinary quality-controlled evaluation from a frozen character-disjoint protocol. This distinction follows the broader document analysis concern that performance under limited labels and unseen writing conditions must be measured rather than inferred from an in-domain split Baloun et al. (2025); Oliveira et al. (2018).

We study the following research questions.

RQ1: How accurately do convolutional and transformer parsers recover the overlapping direction and keypoint representation, and how much performance is lost on unseen character identities?
RQ2: Does a deliberately constrained global registration suppress allowable nuisance transformations while retaining sensitivity to local structural changes?
RQ3: How well do fixed reference-conditioned scores agree with blinded human structural-similarity ratings, and can an interpretable spatial-distribution score improve that association without an unacknowledged post-hoc claim?
RQ4: Which diagnostic evidence can be localized reliably, and where do the current rules or evaluation protocol fail?

Our pipeline separates perception, comparison, and presentation:

$$
\begin{aligned}
\text{overlapping parsing} &\rightarrow \text{constrained registration}\\
&\rightarrow \text{structural comparison}\\
&\rightarrow \text{localized evidence}\\
&\rightarrow \text{text realization.}
\end{aligned}
$$

The parser produces five independently activated direction channels and one keypoint channel. The deployed comparison retains a frozen Dice/IoU/keypoint formula. The paper additionally develops aligned spatial-distribution similarity (ASDS), a compact score based on Jensen–Shannon similarity between polar occupancy, coarse grid occupancy, and horizontal/vertical projection profiles. ASDS is intentionally low-capacity and uses neither character identity nor human ratings at inference. Because its design followed the original rating study, we report development and character-grouped out-of-fold results separately and freeze an entirely disjoint confirmatory pair set before new ratings are collected.

The contributions are:

1. an overlapping six-channel representation and reproducible quality-control protocol for complete-character stroke-structure parsing;
2. a three-architecture, three-seed comparison on both standard and character-disjoint splits, exposing an architecture-dependent trade-off between dense direction masks and sparse keypoints;
3. a constrained registration method evaluated by real-reference controlled perturbations, paired bootstrap intervals, signed-rank tests, and a wider-range negative ablation;
4. an audit of inactive-channel credit and a blinded three-rater study on 150 natural same-character, different-instance pairs;
5. ASDS, an interpretable spatial comparison whose post-rating development status, grouped cross-validation, frozen specification, and independent confirmation protocol are reported explicitly;
6. a diagnostic failure taxonomy that distinguishes scoring limitations, localization-metric limitations, and optional language-model realization.

The intended input is a raster exported from a direct digital writing canvas, not a photograph of paper. Accordingly, the claims concern structural analysis in this modality. They do not cover camera robustness, writer- disjoint generalization, stroke order recovery, or universal aesthetic grading.

2 Related Work

2.1 Chinese handwriting recognition and document segmentation

Large databases such as CASIA-HWDB established modern benchmarks for online and offline handwritten Chinese character recognition Liu et al. (2013). Recognition assigns a character identity, whereas our task requires spatially resolved evidence about directional stroke regions and keypoints. It is therefore closer to document segmentation than to closed-set classification.

U-Net introduced an encoder–decoder architecture with skip connections Ronneberger et al. (2015); DeepLabV3+ combines atrous spatial pyramids with a decoder Chen et al. (2018); and SegFormer uses hierarchical transformer features with a lightweight multilayer-perceptron decoder Xie et al. (2021). Generic document models such as dhSegment illustrate the value of reusable segmentation formulations Oliveira et al. (2018). Recent IJDAR work further shows that label scarcity, source-target mismatch, and the choice of pretraining protocol can materially affect handwritten document segmentation Baloun et al. (2025). We use established architectures to study a different output contract: the five direction labels are independently activated because intersections can support more than one label.

2.2 Computational analysis of Chinese calligraphy

Computational calligraphy research spans handcrafted style descriptors, recognition, generation, and feature extraction. Earlier work represented calligraphic style using local and global shape features Zhang and Nagy (2015); Pengcheng et al. (2017). CalliNet advances style classification with metric learning and triplet supervision Zhang et al. (2026). Other systems extract artistic regions from a calligraphy work relative to a reference Wang et al. (2018), beautify handwritten glyphs toward a Kai-style template Xia and Jin (2009), or evaluate robotic writing through designed aesthetic features Ma and Su (2017).

Large contemporary resources broaden the available calligrapher, script, and page-level annotations. UniCalli, for example, supports column-level calligraphy generation and related representation learning Xu et al. (2026). These resources are valuable for style generation and recognition, but they do not remove the need to validate a practice score against human judgments. Our present study is deliberately narrower: it compares two instances of the same character against a selected reference and reports structural evidence rather than claiming author authentication or style generation.

2.3 Handwritten-character quality assessment

A recent survey divides handwritten Chinese character quality evaluation into rule-based geometric measures, template comparison, and learned methods, while emphasizing the difficulty of defining quality labels and explaining model decisions Chen et al. (2024). Pairwise and Siamese approaches can learn similarity from labelled examples Chen et al. (2025). Multidimensional calligraphy evaluation has also been studied using visual, quality, and aesthetic attributes Wang et al. (2025). These approaches demonstrate that "quality" is not a single observable quantity: legibility, geometric correctness, reference similarity, brushwork, and artistic appeal can disagree.

Our score is consequently scoped as *reference-conditioned structural agreement*. The human study asks raters to judge same-character structural similarity, not beauty or historical fidelity. This distinction is central to the validity claim: a moderate association with structural ratings cannot be relabelled as expert aesthetic calibration.

2.4 Registration and shape comparison

Registration methods range from parametric geometric alignment to learned spatial transformers Zitová and Flusser (2003); Jaderberg et al. (2015). Flexible registration is useful when deformation is a nuisance, but it is hazardous for diagnostic assessment: anisotropic or deformable warping can conceal a proportion or placement error. We therefore restrict alignment to translation, isotropic scale, and small rotation and evaluate that policy with both nuisance and structural perturbations.

Shape contexts describe the spatial distribution of contour points in log-polar bins Belongie et al. (2002); spatial pyramids retain coarse layout that an orderless representation would discard Lazebnik et al. (2006). ASDS follows the same general principle but uses binary-ink occupancy rather than point correspondences. Its distributions are compared with the bounded, symmetric Jensen–Shannon divergence Lin (1991). Polar occupancy captures relative radial/angular organization, a 3×3 grid retains coarse canvas layout, and projection profiles retain row/column mass. The design is simpler than a learned quality regressor and remains directly inspectable.

2.5 Human validation and diagnostic explanation

Automated assessment requires both association with human judgment and a clear account of rater reliability. We report two-way random-effects, absolute-agreement intraclass correlations following established ICC definitions and reporting guidance McGraw and Wong (1996); Koo and Li (2016). Uncertainty for score correlations is estimated by bootstrap resampling at the character level Efron and Tibshirani (1993); paired alignment comparisons additionally use the Wilcoxon signed-rank test Wilcoxon (1945).

Explanation is evaluated separately from score validity. The system first emits normalized evidence—direction channel, missing or extra structure, spatial region, centre offset, scale ratio, and keypoint agreement. A language model may verbalize this evidence, but it does not determine the score. This separation prevents fluent text from being treated as evidence of diagnostic correctness.

3 Method

3.1 Problem formulation

Let I_u be a learner image and I_r an approved reference image of the same target character. The system first predicts overlapping structural masks, then aligns the reference to the learner with a restricted global transform. It returns (i) six masks, (ii) a scalar structural-agreement score, and (iii) localized evidence (Fig. 1). Cross-character scoring is prohibited by the application contract.

3.2 Overlapping six-channel parsing

For an RGB character image I, a parser f_θ predicts

$$P = f_\theta(I) \in [0,1]^{H \times W \times 6}.$$

The fixed channel order is

$$(\mathrm{vec1}, \mathrm{vec2}, \mathrm{vec3}, \mathrm{vec4}, \mathrm{vec5}, \mathrm{keypoint}).$$

The first five channels represent direction-oriented stroke regions; the last represents junction/endpoint evidence. At a crossing, multiple direction labels may be correct at the same pixel. Each channel therefore uses an independent sigmoid rather than a mutually exclusive softmax.

Images are converted to RGB, letterboxed to 512×512, and normalized. Masks are resized by nearest-neighbour interpolation. For direction channel c, the loss combines class-weighted binary cross entropy and soft Dice:

$$\mathcal{L}_{\mathrm{dir}} = \sum_{c=1}^{5} \left(\mathcal{L}_{\mathrm{BCE}}^{(c)} + \mathcal{L}_{\mathrm{Dice}}^{(c)} \right).$$

The sparse keypoint target uses focal loss with $\gamma = 2$ and $\alpha = 0.75$, plus Dice loss. A boundary term derived from the masks is weighted by 0.2:

$$\mathcal{L} = \mathcal{L}_{\mathrm{dir}} + \mathcal{L}_{\mathrm{focal,kp}} + \mathcal{L}_{\mathrm{Dice,kp}} + 0.2\mathcal{L}_{\mathrm{boundary}}.$$

3.3 Versioned reference cache

Approved reference images are parsed once and stored as binary $512 \times 512 \times 6$ arrays. Each cache entry records its reference ID, character, course/style category, parser version, checkpoint SHA-256, thresholds, and channel schema. Caching removes repeated GPU inference for the reference branch. These masks are model-derived references and are never treated as segmentation ground truth.

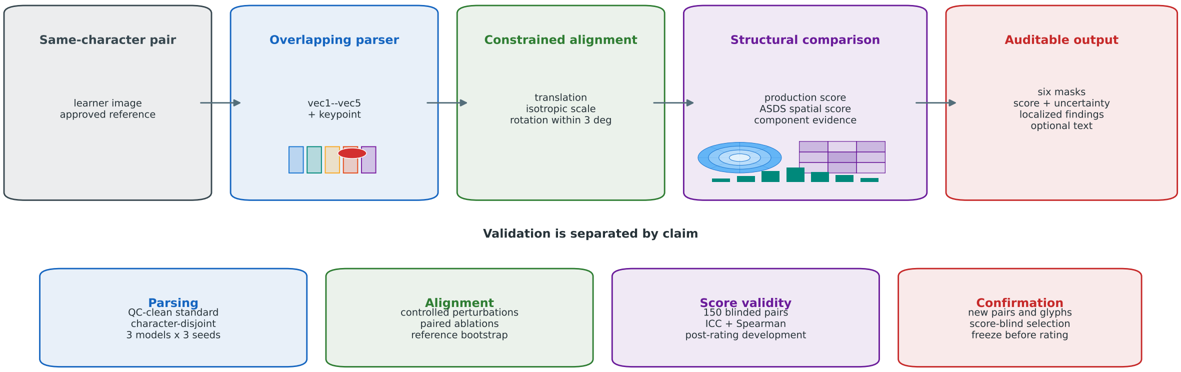

Fig. 1 System and validation overview. The reference and learner are parsed into overlapping direction/keypoint masks, globally aligned under a restricted transform, and compared by the frozen production score and ASDS. Segmentation, alignment, score validity, and confirmation are evaluated by separate protocols so that one successful experiment is not used to support a different claim.

3.4 Constrained similarity registration

Let U and R denote the learner and reference binary stacks, and let

$$M(X) = \bigvee_{c=1}^{5} X_c$$

be the union of direction ink. For each candidate isotropic scale s and rotation ϕ, the transformed reference is translated so that its foreground centroid approaches the learner centroid. The production search is bounded by

$$s \in [0.8, 1.2], \qquad \phi \in [-3^\circ, 3^\circ],$$

and selects the candidate maximizing

$$J\big(M(U), M(T_{s,\phi,t}(R))\big),$$

where J is foreground IoU. Nonuniform scale, shear, thin-plate splines, and deformable warping are disallowed. The restriction encodes the intended invariance: page placement and small global size/rotation changes may be absorbed, whereas local proportion and stroke-region errors should remain.

3.5 Frozen production score

After alignment, let D_c be Dice agreement in direction channel c,

$$\overline{D} = \frac{1}{5}\sum_{c=1}^{5} D_c,$$

let J_{ink} be union-ink IoU, and let $F_{\text{kp}}^{(3)}$ be keypoint-mask F1 with a three-pixel tolerance. The deployed score is

$$S_{\text{prod}} = 100\left(0.55\overline{D} + 0.25J_{\text{ink}} + 0.20F_{\text{kp}}^{(3)}\right).$$

This formula was frozen before the controlled experiments. An audit variant excludes direction channels that are empty in both images and avoids granting perfect keypoint credit when both keypoint masks are empty. The audit score is not retroactively substituted for S_{prod}.

3.6 Aligned spatial-distribution similarity

The human study showed that pixel- and channel-overlap terms incompletely capture perceived global organization. We therefore develop aligned spatial-distribution similarity (ASDS) on the union direction ink after the same frozen registration. The score uses three complementary distributions.

3.6.1 Polar occupancy

For a binary ink mask M, foreground coordinates are centred at their ink centroid. Radius is divided by the 95th percentile of foreground radius and clipped to $[0, 1)$. Each pixel is assigned to one of four radial and eight angular bins, producing a normalized 32-bin histogram $h_p(M)$. Independent centroid and radius normalization make this term resistant to small residual translation and isotropic scale, while retaining angular and radial mass organization.

3.6.2 Coarse grid occupancy

The canonical canvas is partitioned into a 3×3 grid. Normalized foreground counts form $h_g(M)$. Unlike the centroid-normalized polar term, this component retains coarse location and balance on the writing canvas.

3.6.3 Projection profiles

Normalized row sums $h_y(M)$ and column sums $h_x(M)$ describe vertical and horizontal mass allocation. Their similarities are averaged.

For distributions p and q, with $m = (p+q)/2$, we use

$$D_{\mathrm{JS}}(p, q) = \frac{1}{2} D_{\mathrm{KL}}(p\|m) + \frac{1}{2} D_{\mathrm{KL}}(q\|m),$$

with base-2 logarithms, and define

$$\mathrm{JSS}(p, q) = 1 - \sqrt{D_{\mathrm{JS}}(p, q)}.$$

Because $D_{\mathrm{JS}} \in [0, 1]$, $\mathrm{JSS} \in [0, 1]$. The component similarities are

$$s_p = \mathrm{JSS}(h_p(U), h_p(R)),$$
$$s_g = \mathrm{JSS}(h_g(U), h_g(R)),$$

$$s_{\mathrm{proj}} = \frac{1}{2}\big[\,\mathrm{JSS}(h_x(U), h_x(R)) + \mathrm{JSS}(h_y(U), h_y(R))\big].$$

The frozen development specification is

$$S_{\mathrm{ASDS}} = 100\big(0.70 s_p + 0.15 s_g + 0.15 s_{\mathrm{proj}}\big).$$

The development search placed 0.70 weight on the polar term. Rather than retain the more sample-specific unequal secondary weights, we split the remaining 0.30 equally between grid and projection evidence before freezing the confirmatory specification.

The score does not use character ID, writer ID, style ID, or a human rating at inference. Nevertheless, its feature choice and weights were finalized after observing the original 150-pair study. It is therefore a development-stage algorithm until evaluated on a new pair set with no pair or glyph-instance overlap. Features, bins, weights, alignment, and selection policy are frozen before confirmatory ratings begin.

3.7 Evidence extraction and feedback

The system preserves component scores, selected transform, pre-alignment centre distance, ink-area ratio, per-direction agreement, and keypoint agreement. Difference masks are summarized by direction channel, missing reference ink versus extra learner ink, and a 3×3 region set. Connected components of the five direction masks are exposed as direction-classified regions, not as recovered stroke order.

Deterministic feedback ranks centre, scale, keypoint, and local-direction findings. An optional text model receives only this structured contract and may improve fluency; it cannot change the score or invent unsupported brushwork or stroke-order claims.

4 Experimental Protocol

4.1 Legacy segmentation corpus and quality control

The recovered legacy corpus contains 894 sample directories across 43 character IDs. Complete RGB images, five direction masks, a keypoint mask, and a stacked six-channel mask are available for 840 samples from character IDs 0–39. Fifty-four samples from IDs 40–42 lack the character-specific direction mapping and are excluded rather than pseudo-labelled.

File completeness alone was insufficient. A frozen semantic QC procedure identified 12 source-image/GT mismatches and 59 non-canonical members of exact image-and-mask duplicate groups. The exclusion sets do not overlap, leaving 769 unique QC-clean observations.

Duplicate-group membership and mismatch decisions were fixed before the formal model comparison. No exact duplicate group crosses the original train/validation/test assignment.

The original standard assignment contains 600/120/120 samples and becomes 530/119/120 after QC. A second assignment was frozen before evaluation with 28/6/6 disjoint character identities and originally 588/126/126 samples; after the same exclusions it contains 539/114/116 train/validation/test observations. The source assignment, derived QC-clean split, and exclusion contract are content-hashed; their complete SHA-256 values are given in the reproducibility supplement. Writer identity cannot be recovered and writer-disjoint performance is not claimed.

4.2 Training and segmentation evaluation

We compare a 32-base-channel U-Net, ResNet-50 DeepLabV3+, and ImageNet-pretrained SegFormer-B2. Each architecture is run with seeds 20260811, 314159, and 271828 on both split protocols, for 18 formal runs. Inputs are 512×512. Batch size is 8 for U-Net and 4 for the two larger models. Label-safe augmentation includes translation up to 12 pixels, isotropic scale 0.9–1.1, brightness 0.85–1.15, contrast 0.9–1.1, and Gaussian blur with probability 0.15.

All models use AdamW, weight decay 0.01, automatic mixed precision, gradient clipping at 1.0, three warm-up epochs, cosine decay to 10^{-6}, at most 120 epochs, and early stopping with patience 15. U-Net uses learning rate 3×10^{-4}. DeepLabV3+ uses encoder/decoder rates $10^{-4}/10^{-3}$; SegFormer-B2 uses $3 \times 10^{-5}/3 \times 10^{-4}$. Positive BCE weights are estimated from at most 200 training samples and capped at 100. Thresholds are calibrated using the validation set only.

We report five-direction Macro Dice and Macro IoU, per-channel precision/recall/Dice, Boundary F1, strict keypoint F1, and tolerant keypoint F1 at one, three, and five pixels. Test results are summarized as the mean and sample standard deviation across all declared seeds.

4.3 Reference library

The structural experiments use 200 approved references: 100 in the Calli-Tongji Beta Ouyang Xun regular-script category and 100 in the Wang Xizhi running-script category. The reference cache contains six binary masks generated by the frozen SegFormer-B2 v1 checkpoint. It supports seven same-character cross-style pairs but no same-style, different-instance pair. The restricted source images are not redistributed.

4.4 Controlled perturbation benchmark

To isolate scoring and registration from parser error, deterministic perturbations are applied directly to the cached real-reference masks. Nuisance families are translation, rotation, scale up, scale down, and a compound allowed transform. Structural families are terminal deletion, added fragment, local fragment shift, direction dilation, direction erosion, and keypoint shift. Target direction channels and movement axes are selected with stable hashes rather than model errors or manual case choice. A transformed case that risks canvas clipping is marked invalid with a reason and excluded from valid-score summaries.

For each family we report valid N, mean, median, sample standard deviation, 5th and 95th percentiles, 95% reference-bootstrap confidence intervals, invalid fraction, within-reference severity Spearman correlation, and adjacent monotonicity.

4.5 Paired registration ablation

The identical reference/perturbation/severity tuples are evaluated with: (i) no alignment; (ii) the production constrained alignment; and (iii) a preregistered wider search with $s \in [0.6, 1.4]$ and $\phi \in [-12°, 12°]$. Comparisons are paired. Confidence intervals resample references, while Wilcoxon signed-rank tests and rank-biserial effect sizes use per-reference mean paired differences. Family-level comparisons are the primary ablation; perturbation-level p values are unadjusted, descriptive breakdowns.

4.6 Score audit and cross-reference semantics

The score audit counts inactive direction channels and recomputes the coverage-aware candidate described in Sect. 3.5. Qualitative examples are selected by correction quantiles—maximum, upper quartile, median, lower quartile, and near-zero positive—rather than appearance.

For cross-reference semantics, all seven available same-character cross-style pairs and 50 score-independent different-character negative pairs are evaluated. Self-pairs are prohibited. The absent same-character, same-style/different-instance condition is reported as unsupported rather than synthesized.

4.7 Diagnostic benchmark

Perturbation cause supplies deterministic diagnostic ground truth. We report Recall@3, Top-1 cause, direction channel, missing/extra type, exact and overlap grid localization, specificity, and centre-direction wording. Language-model text is disabled. Because the benchmark describes engineered perturbations rather than expert pedagogy, it is a secondary analysis.

4.8 Original blinded human study

Before collecting ratings, 150 natural same-character, different-instance pairs were selected from the 769 QC-clean observations. All 40 character IDs are represented by three or four pairs. Bad images, exact duplicates, and suspected near duplicates were excluded. Pair roles, asset hashes, blinded identifiers, and display order were frozen. Scores and source identifiers were hidden from raters.

Three consenting adult raters independently judged structural similarity on a five-point ordinal scale. Their self-reported backgrounds were heterogeneous: one had one year of calligraphy learning/teaching experience, one reported two years of other relevant experience, and one reported none. We therefore use "human raters", not "three experts". Each rater saw 150 canonical pairs and 15 randomly interspersed hidden repeats.

Validity is measured by Spearman correlation between each score and the three-rater mean on canonical pairs. Confidence intervals use 10,000 bootstrap resamples clustered by target character. Inter-rater reliability uses two-way random-effects, absolute-agreement ICC(2,1) and ICC(2,k). Hidden repeats are excluded from correlation and ICC and are used only for exact agreement, agreement within one point, mean absolute difference, and quadratic weighted kappa.

4.9 Post-rating ASDS development

The original 150 ratings constitute a development set for ASDS, not an independent test. Three fixed distribution families were considered: polar occupancy, 3×3 grid occupancy, and projection profiles. A coarse non-negative simplex search evaluated component weights. The published formula is deliberately regularized: the search-selected polar weight is retained, while the remaining weight is divided equally between grid and projection terms. The full-grid apparent optimum was $(w_g, w_p, w_{\text{proj}}) = (0.225, 0.700, 0.075)$ with $\rho = 0.557$; the frozen 0.15/0.70/0.15 formula gives $\rho = 0.556$. Five-fold cross-validation groups by character ID; each fold's weights are selected on the other characters only. Character-cluster bootstrap intervals are reported for the full frozen formula, grouped out-of-fold predictions, and paired differences from the production and coverage-aware scores.

Component ablations report the polar, grid, and projection terms separately. The combined formula is compared with each component so that any apparent benefit cannot be attributed to an unreported learned regressor. Development component comparisons are exploratory and are not multiplicity-adjusted.

4.10 Independent confirmatory protocol

Before the first confirmatory rating, the algorithm implementation, formula, pair set, asset hashes, rater instructions, analysis script, exclusion rule, and primary endpoint will be frozen. The candidate package contains 100 same-character pairs and 40 ordered reserves. It has zero pair overlap and zero glyph-instance overlap with the original 150-pair study, covers all 40 characters with two or three pairs each, and was selected without reading any production, coverage-aware, ASDS, or human score.

An internal QC reviewer first checks only same-instance suspicion, image quality, accidental duplication, and character correctness. Replacement, if needed, follows the precomputed reserve order without consulting scores. The final CSV will be hashed and renamed `frozen_confirmatory_pairs_v1.csv` before it is sent to raters.

The primary confirmatory endpoint is the Spearman correlation between S_{ASDS} and the mean blinded rating. Secondary endpoints are the production and coverage-aware correlations, paired cluster-bootstrap differences, and rater reliability. No feature, bin, weight, alignment bound, or pair may be changed after rating begins. The submission-ready manuscript must replace [CONFIRMATORY RESULTS] only from this frozen analysis.

5 Results

5.1 RQ1: segmentation and unseen-character generalization

Table 1 shows that, on the QC-clean standard split, DeepLabV3+ obtains the highest direction Macro Dice (0.9630 ± 0.0006), Macro IoU (0.9287 ± 0.0012), and Boundary F1 (0.8443 ± 0.0028). SegFormer-B2 is close in direction overlap but shows greater between-seed variation. U-Net is weaker on direction and boundary metrics yet gives the best strict keypoint F1 (0.7562 ± 0.0038). Thus, dense direction segmentation and sparse keypoint localization favour different architectures.

All models degrade when character identities are held out (Table 2). SegFormer-B2 gives the strongest character-disjoint direction result: Macro Dice 0.8866 ± 0.0008, Macro IoU 0.7977 ± 0.0013, and Boundary F1 0.6851 ± 0.0046. Its Macro-Dice decrease from the standard split is 7.150 percentage points, versus 9.664 for DeepLabV3+ and 9.360 for U-Net. DeepLabV3+ remains second for direction overlap, while U-Net retains the highest strict keypoint F1. SegFormer-B2 is therefore the strongest evaluated model for unseen-character direction parsing, not a universal winner across all six outputs.

The earlier single SegFormer release achieved direction Macro Dice 0.9610 but was trained before semantic QC. No exact duplicate group crosses its standard split, so its result is not explained by exact train–test duplication. Nevertheless, its training population included 12 mismatched and duplicate-weighted observations; all formal comparisons above use the frozen 769-sample QC-clean population.

5.2 RQ2: controlled score behaviour and registration

The family-level results in Table 3 show heterogeneous nuisance robustness. Translation is recovered exactly in all 800 valid observations. Mean absolute penalties are 5.395 for rotation, 13.653 for scale-up, and 16.356 for scale-down. Scale-up and compound transformations have invalid fractions of 49.0% and 54.0%, respectively, because transformed support risks leaving the canvas; those cases are reported rather than silently counted as score failures.

Structural perturbations generally lower the score with severity. Direction dilation and erosion have the largest average penalties, followed by terminal deletion and added fragments. Local fragment shifts and keypoint shifts are penalized more weakly. The benchmark therefore identifies both a desired signal—sensitivity to local structure—and a principal weakness—residual scale sensitivity.

The paired comparison in Table 4 shows that, relative to no alignment, the constrained method reduces nuisance penalties by 30.118 points (reference-bootstrap 95% CI 29.506–30.727) and preserves an additional 11.189 points of structural penalty (10.094–12.322). Both per-reference signed-rank comparisons are nonzero with rank-biserial effect size 1.0.

The wider alignment does not dominate the production range. The current range has a 2.896-point lower nuisance penalty (2.757–3.034), whereas the wider range preserves 0.284 additional points of structural penalty. A wider search therefore fails to improve nuisance suppression and provides only a small structural-sensitivity advantage. The paired family-level breakdown is reported in Supplementary Table 12.

5.3 Score semantics: inactive channels and cross-reference tests

Table 5 shows that 41 of 200 references contain an inactive direction channel: 36 contain one and five contain two; none contains three or more. Empty-channel agreement affects 1,252 valid perturbation rows. Replacing the all-five mean by an active-channel mean changes the score by -0.289 points on average and by as much as -15.402 points. The largest corrections occur when inactive channels

Table 1 QC-clean standard-split segmentation results over three fixed random seeds. Values are mean ± sample standard deviation.

Model	Macro Dice	Macro IoU	Keypoint F1	Boundary F1
U-Net	0.9196 ± 0.0018	0.8515 ± 0.0030	$\mathbf{0.7562 \pm 0.0038}$	0.8009 ± 0.0023
DeepLabV3+	$\mathbf{0.9630 \pm 0.0006}$	$\mathbf{0.9287 \pm 0.0012}$	0.7218 ± 0.0012	$\mathbf{0.8443 \pm 0.0028}$
SegFormer-B2	0.9581 ± 0.0031	0.9196 ± 0.0057	0.7139 ± 0.0046	0.8259 ± 0.0088

Table 2 Character-disjoint test results over three fixed random seeds. Train, validation, and test character identities are disjoint. Values are mean ± sample standard deviation.

Model	Macro Dice	Macro IoU	Keypoint F1	Boundary F1
U-Net	0.8260 ± 0.0161	0.7120 ± 0.0225	$\mathbf{0.6972 \pm 0.0045}$	0.6430 ± 0.0136
DeepLabV3+	0.8664 ± 0.0094	0.7663 ± 0.0146	0.6755 ± 0.0031	0.6728 ± 0.0138
SegFormer-B2	$\mathbf{0.8866 \pm 0.0008}$	$\mathbf{0.7977 \pm 0.0013}$	0.6604 ± 0.0033	$\mathbf{0.6851 \pm 0.0046}$

Fig. 2 Coverage-correction cases selected by correction quantiles rather than visual preference. Purple denotes aligned overlap, blue missing-reference ink, and red extra-user ink.

receive perfect Dice credit while a perturbed active channel degrades.

Figure 2 illustrates the prespecified correction quantiles without selecting examples for visual appeal.

The seven same-character cross-style pairs in Table 6 have mean score 20.239 (bootstrap 95% CI 15.639–25.069), compared with 9.741 (8.584–10.970) for 50 different-character negatives. Cliff's delta is 0.84 Cliff (1993). This is evidence of a same-character reference signal, but the small positive set and missing same-style/different-instance condition prevent a broad style-generalization claim.

5.4 RQ3: blinded human validity and ASDS development

Across 150 canonical pairs, S_{prod} has a positive but modest association with the three-rater mean ($\rho = 0.297$, character-cluster bootstrap 95% CI 0.148–0.441, $p = 0.00023$). The coverage-aware audit is stronger ($\rho = 0.429$, 0.313–0.535, $p < 10^{-7}$). The paired bootstrap difference is 0.132 (0.062–0.197). This result is consistent with the inactive-channel audit but does not, by itself, authorize post-hoc replacement of the production score.

These associations are summarized in Table 7. Canonical ICC(2,1) is 0.448 and ICC(2,k) is 0.709 (Table 8). Across 45 rater-level hidden repeats, exact agreement is 48.9%, agreement within one scale point is 93.3%, mean absolute difference is 0.578, and pooled quadratic weighted kappa is 0.634. Individual production-score correlations are positive for all raters and range from 0.174 to 0.350. Average ratings are therefore more reliable than a single rating, but meaningful subjectivity remains.

ASDS raises the development-set association to $\rho = 0.556$ (character-cluster bootstrap 95% CI

Table 3 Controlled perturbation summary. Nuisance rows report absolute score drop; structural rows report signed drop from identity. Full distribution and severity statistics are in Supplementary Table 11.

Perturbation	Valid N	Mean drop	95% CI	Invalid fraction	Monotonic
Translation	800	0.000	[0.000, 0.000]	0.000	1.000
Rotation	520	5.395	[5.079, 5.747]	0.133	0.647
Scale up	408	13.653	[12.964, 14.420]	0.490	0.271
Scale down	800	16.356	[15.670, 17.074]	0.000	0.500
Compound allowed transform	276	8.143	[7.621, 8.717]	0.540	0.466
Terminal deletion	800	19.757	[17.387, 22.087]	0.000	0.992
Added direction fragment	800	11.398	[9.934, 12.929]	0.000	0.987
Local fragment shift	800	4.310	[3.886, 4.772]	0.000	0.985
Direction dilation	800	24.034	[22.180, 25.955]	0.000	1.000
Direction erosion	800	22.225	[20.269, 24.344]	0.000	0.997
Keypoint shift	800	7.468	[7.294, 7.641]	0.000	1.000

Table 4 Paired alignment comparison. Positive benefit differences favour the current constrained alignment. Wilcoxon tests use per-reference mean paired differences.

Comparison	Family	Refs	Ours	Comparator	Benefit 95% CI	p	RBC
Ours–none	nuisance	200	8.455	38.573	[29.506, 30.727]	< 0.001	1.000
Ours–none	structural	200	14.865	3.677	[10.094, 12.322]	< 0.001	1.000
Ours–wide	nuisance	200	8.455	11.351	[2.757, 3.034]	< 0.001	1.000
Ours–wide	structural	200	14.865	15.149	[-0.355, -0.213]	< 0.001	-0.683

Table 5 Inactive direction channels among 200 references.

Inactive channels	References	Fraction
0	159	0.795
1	36	0.180
2	5	0.025

0.454–0.651). Predictions produced by character-grouped five-fold weight selection retain $\rho = 0.540$ (0.432–0.639). The paired bootstrap interval for ASDS minus production is 0.110–0.410, and that for ASDS minus the coverage-aware audit is 0.017–0.239. For grouped out-of-fold predictions, the difference from coverage-aware scoring is less certain (-0.003–0.224).

The component analysis in Table 9 identifies the polar term as the strongest individual spatial descriptor ($\rho = 0.547$); grid and projection similarities reach 0.447 and 0.445. Thus, most ASDS association comes from radial/angular ink organization, while the two coarse canvas terms provide a small complementary contribution. The combined score is not claimed to be significantly better than the polar term alone.

These results show that human ratings are more closely related to spatial ink organization than to the deployed channel/pixel overlap alone. They are not an independent validation of ASDS, because the same 150 ratings informed its development. The independent result remains **[CONFIRMATORY ASDS RHO, 95% CI, AND PAIRED DIFFERENCES]**.

Figure 3 visualizes the stronger monotonic organization of ASDS while preserving the substantial dispersion that motivates independent confirmation.

5.5 RQ4: diagnostic feedback

Table 10 separates the dominant localization failure types. Required Recall@3 is 0.701, whereas exact grid-region localization is 0.109. Among exact-region failures, 53.9% occur because the perturbation truth spans multiple cells while the metric requires one exact cell, 24.6% involve an incorrect direction channel, and 20.1% contain no local finding in the top three. Wrong missing/extra type and residual registration shifts are uncommon. The main limitation is therefore a mixture of localization representation and direction attribution rather than wording. This motivates a future set- or pixel-overlap localization metric, not further tuning of feedback rules on the same cases.

Table 6 Cross-reference structural agreement.

Pair type	N	Mean	Median	Mean 95% CI
Same character, cross style	7	20.239	20.272	[15.639, 25.069]
Different-character negative	50	9.741	8.869	[8.584, 10.970]

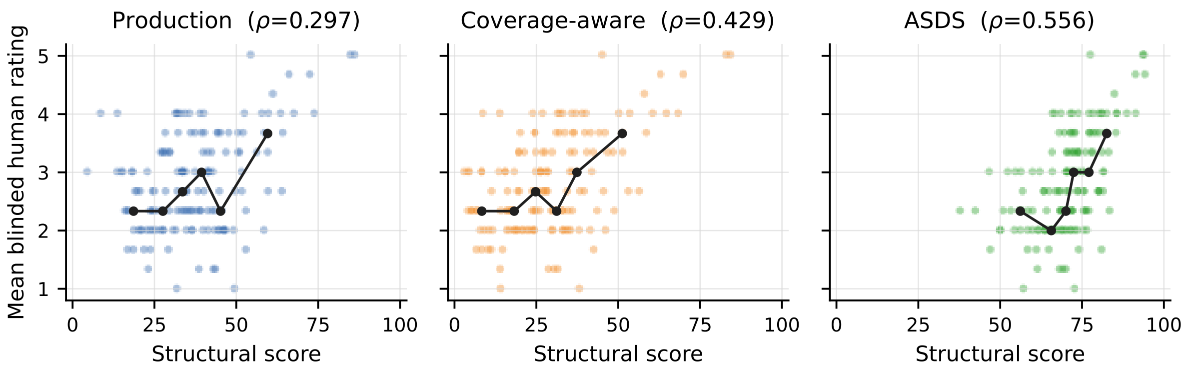

Fig. 3 Development-stage association between blinded mean structural rating and the production, coverage-aware, and ASDS scores for 150 natural same-character pairs. Points are individual pairs; lines connect score-bin medians only as a descriptive guide. ASDS was designed after these ratings, so this figure is not an independent validation.

Table 7 Association between the structural scores and the mean rating of three blinded human raters across 150 canonical pairs. Confidence intervals use 10,000 bootstrap resamples clustered by character.

Score	ρ	95% CI	p
Production	0.297	[0.148, 0.441]	2.28×10^{-4}
Coverage-aware	0.429	[0.313, 0.535]	4.32×10^{-8}

Table 8 Human-rater reliability. ICC uses only the 150 canonical presentations. Repeat statistics use the 15 hidden repeats per rater.

Rater	Exact	Within ± 1	MAD	QWK
E01	0.600	1.000	0.400	0.559
E02	0.467	1.000	0.533	0.721
E03	0.400	0.800	0.800	0.591
POOLED	0.489	0.933	0.578	0.634

Canonical ICC(2,1)=0.448 and ICC(2,k)=0.709 for 150 pairs and 3 raters.

6 Discussion

6.1 What the experiments establish

The first result is that model selection depends on the generalization target. DeepLabV3+ is strongest on the QC-clean standard split, but SegFormer-B2 has the best character-disjoint direction overlap and the smallest decline between protocols. U-Net remains strongest on strict key-point localization. This architecture-by-output interaction argues against selecting a parser from one aggregate metric. It also explains why the preliminary standard-split SegFormer result was insufficient evidence for unseen-character performance.

The second result is that restricted registration is not merely an implementation detail. Without alignment, allowable global geometry dominates the score. The production range sharply improves nuisance behaviour and retains structural penalties. The wider range fails to improve nuisance suppression, providing a useful negative result: more transform capacity is not automatically better for diagnostic comparison. Scale remains the main weakness, but further adjustment on the same perturbation suite would risk benchmark-specific overfitting.

The third result is semantic. Both-empty direction channels can contribute perfect Dice even though no active structure is compared. The average effect is small, but a limited set of cases receives a large correction. The stronger human association of the coverage-aware audit supports

Table 9 Development-stage ASDS component ablation and association with mean human structural rating. Confidence intervals use 10,000 character-cluster bootstrap resamples. ASDS was developed after the original ratings and is not an independent result.

Score or component	Spearman ρ	95% CI
Grid JS similarity only	0.447	[0.303, 0.572]
Projection JS similarity only	0.445	[0.316, 0.564]
Polar JS similarity only	0.547	[0.443, 0.645]
ASDS frozen development formula	0.556	[0.454, 0.651]
ASDS character-grouped OOF	0.540	[0.432, 0.639]

Table 10 Primary taxonomy of exact-region diagnostic failures.

Failure type	Count	Fraction
Multi-region/boundary truth	1922	0.539
Wrong direction channel	876	0.246
Local finding absent from Top-3	718	0.201
Wrong missing/extra type	38	0.011
Alignment residual shift	11	0.003

this concern. ASDS extends the finding: coarse radial, angular, and projection distributions track the human structural rating better than the production score in the development study. This is plausible because raters can perceive global proportion and mass distribution even when exact channel overlap is modest.

6.2 Why ASDS remains a development result

It would be statistically inappropriate to present $\rho = 0.556$ as a final generalization result. The spatial representation and weights were finalized after the original ratings were available. Character-grouped out-of-fold predictions reduce weight-selection optimism, but they cannot erase the fact that feature families were proposed after observing the same study.

We therefore preserve three layers of evidence: (i) the originally frozen production result; (ii) a transparent post-rating development analysis, including all considered feature families and grouped cross-validation; and (iii) a fully disjoint confirmatory study whose pair identities and glyph instances do not overlap the development set. Only the third layer can support a claim that ASDS improves human association on new material. This separation is less visually attractive than reporting only the best ρ, but it is necessary for reproducible score development.

ASDS is also intentionally small. Its three bounded components are directly inspectable and can be computed on CPU. Component ablations test whether the combined score adds value beyond its dominant polar term. If the independent study does not confirm the improvement, the production score remains deployed and ASDS should be reported as a negative development result rather than retuned on the confirmatory data.

6.3 Interpretation of human agreement

The average-rating ICC is substantially higher than the single-rating ICC. This supports averaging multiple judgments for the validation endpoint, while also showing that the task is not objective in the sense of a segmentation mask. The rater pool is small and heterogeneous. One rater had limited calligraphy teaching/learning experience, but the group is not a panel of senior calligraphy experts. The results justify the phrase "human-validated structural agreement"; they do not justify "expert aesthetic score".

A stronger follow-up should recruit at least two experienced calligraphy educators, record background through a fixed questionnaire, and analyse whether professional and non-professional rankings differ. Such a study would extend the construct from general structural similarity toward pedagogical quality, but it should not be retroactively substituted for the current task.

6.4 Diagnostic and application implications

The low exact-region result initially appears to indicate poor feedback. Failure analysis shows that more than half of these errors arise from a single-cell metric applied to multi-cell perturbations. The next diagnostic benchmark should store affected support as a region set or pixel mask and use overlap-based scoring. Wrong direction attribution remains a genuine model or evidence-extraction failure and warrants separate study. Optional language generation should be evaluated only after the structured evidence contract is validated; fluent text cannot repair an incorrect direction or region.

The target platform receives direct digital writing. This modality is aligned with the held-out legacy images, but it does not imply robustness to phone photographs, paper texture, illumination, or perspective. The SegFormer-B2 parser benefits from a GPU for interactive latency; reference masks are precomputed, and registration, ASDS, deterministic feedback, and language-API calls do not require a dedicated training-class GPU. Distillation and quantized edge deployment are engineering extensions rather than claims of this study.

6.5 Limitations and threats to validity

The corpus contains only 40 fully labelled character identities. Writer identity and verified style provenance are unavailable, so character-disjoint evaluation is not writer-disjoint evaluation. The standard and character-disjoint results come from one recovered data source, not an external corpus. Exact duplicate leakage has been audited, but unobserved collection-specific regularities may remain.

The structural perturbation experiments operate on cached reference masks and isolate score behaviour from parser errors. This is a deliberate internal validity choice, but an end-to-end error can combine segmentation, registration, and scoring failures. The reference library contains only two style categories, seven same-character cross-style pairs, and no same-style different-instance pairs. Broader course coverage cannot be inferred from these comparisons.

The original human study contains 150 pairs and three raters. The confirmatory candidate set is drawn from the same recovered corpus, although it is disjoint at both pair and glyph-instance levels. Consequently, successful confirmation would establish within-corpus generalization, not external-dataset validity. Finally, ASDS uses the union of five direction masks and can ignore a direction-label swap that leaves overall ink distribution unchanged. It should therefore complement, not erase, the channel-level evidence returned to users.

7 Conclusion

We presented a reference-conditioned pipeline for low-resource Chinese calligraphy analysis that combines overlapping direction/keypoint parsing, restricted global registration, interpretable structural comparison, and localized evidence. Across three fixed seeds, DeepLabV3+ is strongest on the QC-clean standard split, while SegFormer-B2 gives the best character-disjoint direction result and U-Net gives the best strict keypoint result. Controlled perturbations and paired ablations show that the constrained registration suppresses nuisance geometry without freely warping away local structural changes.

The human study also changes the interpretation of the score. The deployed formula has a modest positive association with blinded structural-similarity ratings, and a coverage-aware audit is stronger. The proposed ASDS score, which compares polar occupancy, coarse grid occupancy, and projection profiles, raises development-stage Spearman correlation from 0.297 to 0.556 and retains 0.540 under character-grouped out-of-fold prediction. Because ASDS was developed after observing the original ratings, these values are exploratory. A pair- and instance-disjoint confirmatory study is required before replacement of the production score.

The resulting system should be viewed as an auditable structural-analysis tool, not a universal aesthetic grader. Its strongest scientific contribution is the joint evaluation discipline: quality-controlled and character-disjoint parsing, controlled registration tests, semantic score audits, blinded human association, explicit post-hoc development, and a prospectively frozen confirmation protocol.

Supplementary statistical tables

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Conflict of interest

The authors declare no competing interests.

Ethics approval and consent to participate

The structural-similarity study involved adult volunteers, stored ratings under evaluator codes, and did not collect sensitive personal information. Before submission, the authors will insert the institution-specific ethics approval, exemption, or review statement and the applicable consent wording. No approval or exemption is claimed in this draft.

Consent for publication

Not applicable, subject to confirmation by the authors before submission.

Data availability

The recovered segmentation corpus and the Calli-Tongji-derived reference images are subject to their original ownership and licence constraints. Reproducibility artifacts provide frozen identifiers, hashes, split contracts, statistics, and generation scripts without redistributing restricted source images. The exact public/private release statement will be supplied before submission.

Code availability

[ANONYMIZED REPOSITORY DURING REVIEW OR PUBLIC ARCHIVAL URL]

Author contributions

[CRediT CONTRIBUTION STATEMENT]

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Table 11 Full per-perturbation score-drop statistics. Nuisance rows use absolute drop; structural rows use signed drop from the identity score. Confidence intervals use reference-level bootstrap resampling.

Perturbation	N	Mean	Median	SD	95% CI	P05–P95	Invalid	ρ_s	Monotonic
Translation	800	0.000	0.000	0.000	[0.000, 0.000]	[0.000, 0.000]	0.000	–	1.000
Rotation	520	5.395	4.890	2.335	[5.079, 5.747]	[3.045, 9.024]	0.133	-0.521	0.647
Scale up	408	13.653	12.775	4.556	[12.964, 14.420]	[8.030, 22.745]	0.490	0.812	0.271
Scale down	800	16.356	15.412	5.225	[15.670, 17.074]	[10.051, 26.419]	0.000	-0.662	0.500
Compound transform	276	8.143	7.298	3.468	[7.621, 8.717]	[4.295, 15.464]	0.540	0.397	0.466
Terminal deletion	800	19.757	11.113	20.424	[17.387, 22.087]	[0.403, 62.702]	0.000	-0.993	0.992
Added direction fragment	800	11.398	6.935	12.736	[9.934, 12.929]	[0.277, 41.357]	0.000	-0.988	0.987
Local fragment shift	800	4.310	2.537	4.220	[3.886, 4.772]	[0.547, 12.677]	0.000	-0.985	0.985
Direction dilation	800	24.034	17.866	18.673	[22.180, 25.955]	[3.336, 62.536]	0.000	-1.000	1.000
Direction erosion	800	22.225	15.425	18.048	[20.269, 24.344]	[3.550, 60.381]	0.000	-0.992	0.997
Keypoint shift	800	7.468	4.570	6.971	[7.294, 7.641]	[0.000, 18.232]	0.000	-1.000	1.000

Table 12 Paired alignment comparisons by perturbation family. Positive benefit differences favour the current constrained alignment.

Comparison	Perturbation	Refs	Ours	Comparator	Benefit 95% CI	p	RBC
Ours–none	compound transform	156	8.143	43.947	[34.353, 37.172]	< 0.001	1.000
Ours–none	terminal deletion	200	19.757	2.135	[15.447, 19.802]	< 0.001	1.000
Ours–none	direction dilation	200	24.034	4.624	[17.685, 21.234]	< 0.001	1.000
Ours–none	direction erosion	200	22.225	4.641	[15.636, 19.565]	< 0.001	1.000
Ours–none	added fragment	200	11.398	0.923	[9.021, 11.985]	< 0.001	1.000
Ours–none	rotation	180	5.395	15.927	[10.176, 10.871]	< 0.001	1.000
Ours–none	scale down	200	16.356	48.355	[31.504, 32.517]	< 0.001	1.000
Ours–none	scale up	170	13.653	35.869	[20.706, 23.708]	< 0.001	0.988
Ours–none	translation	200	0.000	43.034	[42.054, 44.005]	< 0.001	1.000
Ours–none	keypoint shift	200	7.468	7.468	[0.000, 0.000]	1.000	0.000
Ours–none	local fragment shift	200	4.310	2.270	[1.684, 2.427]	< 0.001	1.000
Ours–wide	compound transform	156	8.143	22.951	[14.336, 15.312]	< 0.001	1.000
Ours–wide	terminal deletion	200	19.757	20.231	[-0.639, -0.317]	< 0.001	-0.696
Ours–wide	direction dilation	200	24.034	24.605	[-0.751, -0.410]	< 0.001	-0.623
Ours–wide	direction erosion	200	22.225	22.730	[-0.682, -0.341]	< 0.001	-0.577
Ours–wide	added fragment	200	11.398	11.560	[-0.280, -0.054]	0.005	-0.407
Ours–wide	rotation	180	5.395	12.019	[6.344, 6.908]	< 0.001	1.000
Ours–wide	scale down	200	16.356	16.861	[0.439, 0.571]	< 0.001	0.919
Ours–wide	scale up	170	13.653	14.107	[0.377, 0.533]	< 0.001	0.841
Ours–wide	translation	200	0.000	0.000	[0.000, 0.000]	1.000	0.000
Ours–wide	keypoint shift	200	7.468	7.468	[0.000, 0.000]	1.000	0.000
Ours–wide	local fragment shift	200	4.310	4.302	[-0.006, 0.024]	0.263	0.444

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